WORKED SOLUTION

STEP I 1999, Question 2

Pure • Locus and Centroid

STEP I 1999, Question 2

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/99-s1-q2

The key idea

Write down the condition, expand, and you get x^2 and y^2 each with coefficient 3 and no xy term: that is the signature of a circle. Completing the square shows where its centre is, and the centroid appears naturally because $x_1 + x_2 + x_3$ is exactly what turns up in the linear terms. The constraint on a^2 then drops out of asking when the "radius squared" is allowed to be negative.

Solution

The locus

Let $P = (x, y)$ be a point of the locus, and write $A_i = (x_i, y_i)$ for $i = 1, 2, 3$. The condition is

$$\sum_{i=1}^{3} \left[(x - x_i)^2 + (y - y_i)^2 \right] = a^2.$$

Expanding, and using $x_1 + x_2 + x_3 = 3\bar{x}$ and $y_1 + y_2 + y_3 = 3\bar{y}$,

$$3x^2 - 6\bar{x}\,x + 3y^2 - 6\bar{y}\,y + \sum_i (x_i^2 + y_i^2) = a^2.$$

This is the equation of the locus. To see what it is, divide by 3 and complete the square:

$$(x - \bar{x})^2 + (y - \bar{y})^2 = \frac{a^2 - \sum_i (x_i^2 + y_i^2) + 3\bar{x}^2 + 3\bar{y}^2}{3}.$$

The numerator tidies up nicely. Since $\sum_i x_i = 3\bar{x}$,

$$\sum_i (x_i - \bar{x})^2 = \sum_i x_i^2 - 2\bar{x} \cdot 3\bar{x} + 3\bar{x}^2 = \sum_i x_i^2 - 3\bar{x}^2,$$

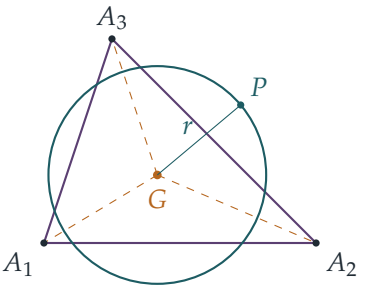
and similarly for y. So if we write

$$S = \sum_{i=1}^{3} \left[(x_i - \bar{x})^2 + (y_i - \bar{y})^2 \right]$$

for the sum of the squares of the distances of the three points from their centroid $G = (\bar{x}, \bar{y})$, the locus is

$$\boxed{(x-\bar{x})^2 + (y-\bar{y})^2 = \frac{a^2 - S}{3}}.$$

Interpretation. When $a^2 > S$ this is a circle with centre the centroid G of the three points and radius $\sqrt{(a^2 - S)/3}$. As a^2 varies we get a family of concentric circles about G. In the boundary case $a^2 = S$ the "circle" shrinks to the single point G.



Why a^2 cannot be less than S

> **Ask yourself** What does it mean for the right-hand side of the boxed equation to be negative?

The left-hand side $(x-\bar{x})^2 + (y-\bar{y})^2$ is a sum of squares of real numbers, so it is never negative. If $a^2 < S$, the right-hand side is negative and no real point (x, y) satisfies the equation: the locus is empty. So for a point to move as described at all, we need $a^2 \geq S$.

It is illuminating to read the same working the other way round. What we actually proved is the identity, valid for every point P,

$$PA_1^2 + PA_2^2 + PA_3^2 = 3\,PG^2 + S.$$

Since $PG^2 \geq 0$, the sum $PA_1^2 + PA_2^2 + PA_3^2$ is always at least S, with equality only at $P = G$. In words: the centroid is the point that minimises the sum of the squared distances to the three points, and the minimum value is S. A constant sum a^2 smaller than this minimum is impossible.

Checks

- Take $A_1 = (0,0)$, $A_2 = (4,0)$, $A_3 = (1,3)$, so $G = (\frac{5}{3}, 1)$ and $S = \frac{34}{9} + \frac{58}{9} + \frac{40}{9} = \frac{44}{3}$. At $P = G$ the sum of squared distances is exactly $\frac{44}{3} \approx 14.67$; at $P = (0,0)$ it is $0 + 16 + 10 = 26$, and $3PG^2 + S = 3 \cdot \frac{34}{9} + \frac{44}{3} = 26$, as the identity predicts.
- Another useful form: $S = \frac{1}{3}(A_1A_2^2 + A_2A_3^2 + A_3A_1^2)$. In the example the squared sides are 16, 18 and 10, giving $\frac{44}{3}$ again.

- We checked the identity exactly, with fractions, on 2000 random triples of points, and confirmed numerically that points on the circle give the sum a^2 and that the minimum of the sum is S, attained at G.

Common slips

- Leaving the radius as $\sqrt{\left(a^2 - \sum(x_i^2 + y_i^2)\right)/3 + \bar{x}^2 + \bar{y}^2}$ without recognising it in terms of S; the final part then becomes very hard to see.
- Forgetting the degenerate case $a^2 = S$, where the locus is a single point, not a circle.

Beyond STEP

The identity $\sum PA_i^2 = 3PG^2 + \sum GA_i^2$ is a form of the parallel axis theorem for moments of inertia, and in statistics it is the decomposition of a sum of squares into "between" and "within" parts: the mean is the value that minimises the sum of squared deviations, which is the whole reason least squares fitting works. In linear algebra it is Pythagoras' theorem in disguise, because the vector from G to P is orthogonal (in the right sense) to the deviations $A_i - G$, which sum to zero. Strang's *Introduction to Linear Algebra* develops this projection viewpoint of least squares.

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